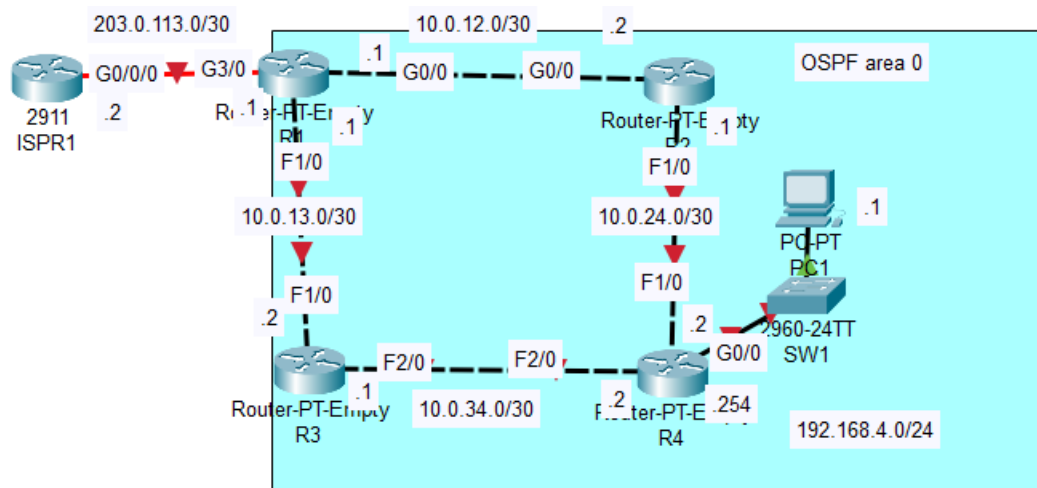




Topology

1. Configure the appropriate hostnames and IP addresses on each device. Enable router interfaces. (You don't have to configure ISPR1)

```
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R1
R1(config)#int g0/0
R1(config-if)#ip add 10.0.12.1 255.255.255.252
R1(config-if)#no shutdown

R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

R1(config-if)#int f1/0
R1(config-if)#ip add 10.0.13.1 255.255.255.252
R1(config-if)#no shutdown

R1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet1/0, changed state to up

R1(config-if)#int g3/0
R1(config-if)#ip add 203.0.113.1 255.255.255.252
R1(config-if)#no shut

R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet3/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet3/0, changed state to up
```

```
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R2
R2(config)#int g0/0
R2(config-if)#ip add 10.0.12.2 255.255.255.252
R2(config-if)#no shut

R2(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

R2(config-if)#int f1/0
R2(config-if)#ip add 10.0.24.1 255.255.255.252
R2(config-if)#no shut

R2(config-if)#
%LINK-5-CHANGED: Interface FastEthernet1/0, changed state to up

Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R3
R3(config)#int f1/0
R3(config-if)#ip add 10.0.13.2 255.255.255.252
R3(config-if)#no shut

R3(config-if)#
%LINK-5-CHANGED: Interface FastEthernet1/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0, changed state to up

R3(config-if)#int f2/0
R3(config-if)#ip add 10.0.34.1 255.255.255.252
R3(config-if)#no shut

R3(config-if)#
%LINK-5-CHANGED: Interface FastEthernet2/0, changed state to up
```

```

Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R4
R4(config)#ip add 10.0.34.2 255.255.255.252
      ^
% Invalid input detected at '^' marker.

R4(config)#int f2/0
R4(config-if)#ip add 10.0.34.2 255.255.255.252
R4(config-if)#no shut

R4(config-if)#
%LINK-5-CHANGED: Interface FastEthernet2/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet2/0, changed state to up

R4(config-if)#int f1/0
R4(config-if)#ip add 10.0.24.2 255.255.255.252
R4(config-if)#no shut

R4(config-if)#
%LINK-5-CHANGED: Interface FastEthernet1/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0, changed state to up

R4(config-if)#int g0/0
R4(config-if)#ip add 192.168.4.2 255.255.255.0
R4(config-if)#no shut

R4(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

```

2. Configure a loopback interface on each router (1.1.1.1/32 for R1, 2.2.2.2/32 for R2, etc.)

```

R1(config)#int 10

R1(config-if)#
%LINK-5-CHANGED: Interface Loopback0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to up

R1(config-if)#ip add 1.1.1.1 255.255.255.255
R2(config)#int 10

R2(config-if)#
%LINK-5-CHANGED: Interface Loopback0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to up

R2(config-if)#ip add 2.2.2.2 255.255.255.255
R3(config)#int 10

R3(config-if)#
%LINK-5-CHANGED: Interface Loopback0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to up

R3(config-if)#ip add 3.3.3.3 255.255.255.255

```

```
R4(config)#int 10
```

```
R4(config-if)#
```

```
%LINK-5-CHANGED: Interface Loopback0, changed state to up
```

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to up
```

```
R4(config-if)#ip add 4.4.4.4 255.255.255.255
```

3. Enable OSPF directly on each interface of the routers. Configure passive interfaces as appropriate.

```
R1(config)#int range g0/0, f1/0, 10
```

```
R1(config-if-range)#ip ospf 1 area 0
```

```
R1(config-if-range)#do sh ip protocols
```

```
Routing Protocol is "ospf 1"
```

```
Outgoing update filter list for all interfaces is not set
```

```
Incoming update filter list for all interfaces is not set
```

```
Router ID 1.1.1.1
```

```
Number of areas in this router is 1. 1 normal 0 stub 0 nssa
```

```
Maximum path: 4
```

```
Routing for Networks:
```

```
Routing Information Sources:
```

Gateway	Distance	Last Update
1.1.1.1	110	00:03:11

```
Distance: (default is 110)
```

```
R1(config-if-range)#router ospf 1
```

```
R1(config-router)#passive-interface 10
```

```
R2(config)#int range f1/0, g0/0, 10
```

```
R2(config-if-range)#ip ospf 1 area 0
```

```
R2(config-if-range)#router ospf 1
```

```
R2(config-router)#passive-interface 10
```

```
R3(config)#int range f1/0, f2/0
```

```
R3(config-if-range)#ip ospf 1 area 0
```

```
R3(config-if-range)#router ospf 1
```

```
R3(config-router)#passive-inter
```

```
23:29:04: %OSPF-5-ADJCHG: Process 1, Nbr 1.1.1.1 on FastEthernet1/0 from LOADING to FULL, Loading Done
```

```
face 10
```

```
R3(config-router)#int range 10
```

```
R3(config-if-range)#ip ospf 1 area 0
```

```
R4(config-if)#ip add 4.4.4.4 255.255.255.255
```

```
R4(config-if)#exit
```

```
R4(config)#int range f1/0, f2/0, 10
```

```
R4(config-if-range)#ip ospf 1 area 0
```

```
R4(config-if-range)#router ospf 1
```

```
R4(config-router)#passive-interface
```

```
23:30:14: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on FastEthernet1/0 from LOADING to FULL, Loading Done
```

```
10
```

4. Configure the reference bandwidth on each router so a FastEthernet interface has a cost of 100.

```

R1(config-router)#auto-cost reference-bandwidth 10000
% OSPF: Reference bandwidth is changed.
    Please ensure reference bandwidth is consistent across all routers.
R1(config-router)#do sh ip ospf int

GigabitEthernet0/0 is up, line protocol is up
  Internet address is 10.0.12.1/30, Area 0
  Process ID 1, Router ID 1.1.1.1, Network Type BROADCAST, Cost: 10
  Transmit Delay is 1 sec, State DR, Priority 1
  Designated Router (ID) 1.1.1.1, Interface address 10.0.12.1
  Backup Designated Router (ID) 2.2.2.2, Interface address 10.0.12.2
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    Hello due in 00:00:05
  Index 1/1, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 1, Adjacent neighbor count is 1
    Adjacent with neighbor 2.2.2.2 (Backup Designated Router)
  Suppress hello for 0 neighbor(s)
FastEthernet1/0 is up, line protocol is up
  Internet address is 10.0.13.1/30, Area 0
  Process ID 1, Router ID 1.1.1.1, Network Type BROADCAST, Cost: 100
  Transmit Delay is 1 sec, State DR, Priority 1
  Designated Router (ID) 1.1.1.1, Interface address 10.0.13.1
  Backup Designated Router (ID) 3.3.3.3, Interface address 10.0.13.2
--More--

R2(config-router)#auto-cost reference-bandwidth 10000
% OSPF: Reference bandwidth is changed.
    Please ensure reference bandwidth is consistent across all routers.
R3(config-router)#auto-cost reference-bandwidth 10000
% OSPF: Reference bandwidth is changed.
    Please ensure reference bandwidth is consistent across all routers.
R4(config-router)#auto-cost reference-bandwidth 10000
% OSPF: Reference bandwidth is changed.
    Please ensure reference bandwidth is consistent across all routers.

```

5. Configure R1 as an ASBR that advertises a default route in to the OSPF domain.

```

R1(config-router)#default-information originate
R1(config-router)#exit
R1(config)#ip route 0.0.0.0 0.0.0.0 203.0.113.2

```

6. Check the routing tables of R4. What default route(s) were added? (watch the video for a brief explanation)

R2 and R3 routes

7. Use Simulation mode to view the OSPF Hello messages being sent by the routers. What fields are included in the Hello message?

The image displays a Cisco Packet Tracer simulation environment. The main window shows a network topology with several routers (R1, R2, R3, R4) and a switch (SW1). The routers are interconnected via their FastEthernet interfaces. The IP addresses for the interfaces are as follows:

- R1: 10.0.12.0/30 (G0/0/0), 10.0.13.0/30 (F1/0), 10.0.14.0/30 (F2/0)
- R2: 10.0.12.0/30 (G0/0/0), 10.0.13.0/30 (F1/0), 10.0.14.0/30 (F2/0)
- R3: 10.0.13.0/30 (F1/0), 10.0.14.0/30 (F2/0), 10.0.15.0/30 (F3/0)
- R4: 10.0.12.0/30 (G0/0/0), 10.0.13.0/30 (F1/0), 10.0.14.0/30 (F2/0)

The simulation panel on the right shows the Event List with the following entries:

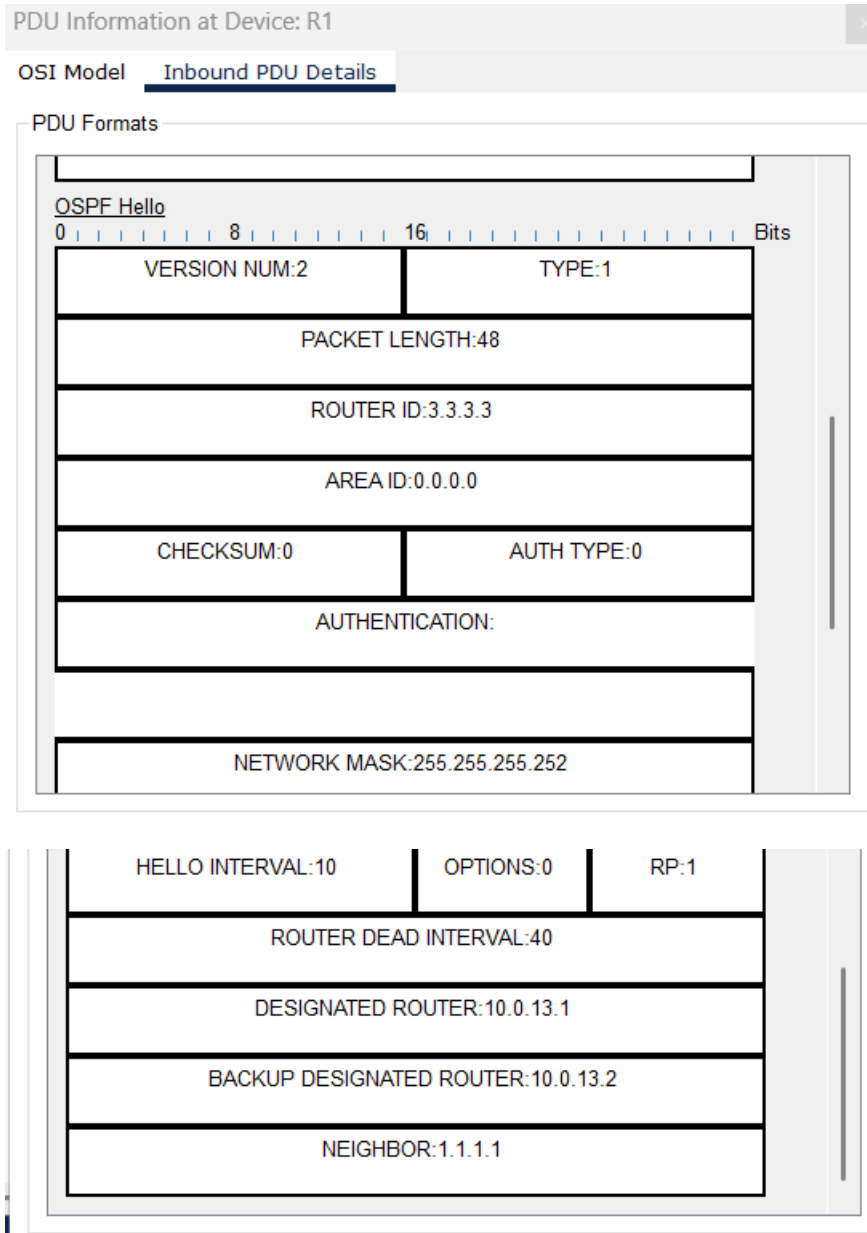
Vis.	Time(sec)	Last Device	At Device	Type
	2.971	SW1	PC1	STP
	2.971	SW1	R4	STP
	3.316	--	R3	OSPF
	3.317	R3	R4	OSPF
	3.325	--	R3	OSPF
	3.326	R3	R1	OSPF

The PDU Information at Device: R1 window shows the following details:

- At Device:** R1
- Source:** R3
- Destination:** 224.0.0.5
- In Layers:** Layer7, Layer6, Layer5, Layer4, Layer3, Layer2, Layer1
- Out Layers:** Layer7, Layer6, Layer5, Layer4, Layer3, Layer2, Layer1
- Layer 3:** IP Header Src. IP: 10.0.13.2, Dest. IP: 224.0.0.5 OSPF HELLO
- Layer 2:** Ethernet II Header 0030.A30A.C407 >> 0100.5E00.0005
- Layer 1:** Port FastEthernet1/0

The bottom of the image shows the Play Controls and a list of devices in the simulation.

224.0.0.5 – multicast address for all OSPF routers, and IP address that routers send hello messages to



Version Num 2 is used – the type usually used for IPv4

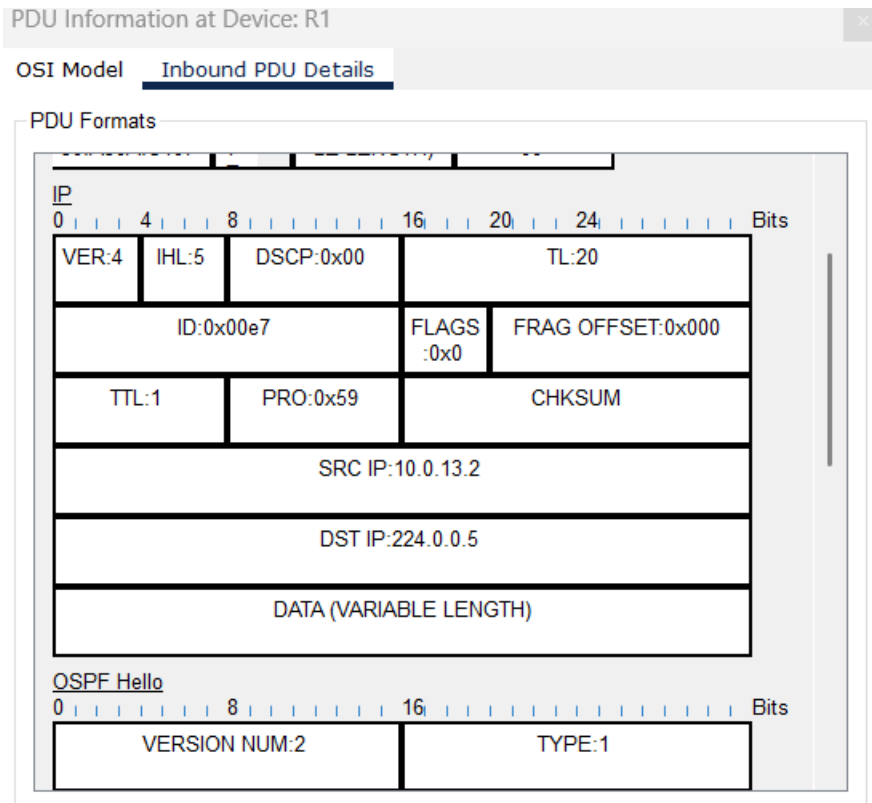
Type 1 – A Type 1 OSPF message, a Hello message

Router ID 3.3.3.3

Area ID 0.0.0.0 – Dotted decimal translation of Area '0'

Hello Interval of 10s; Dead Interval of 40s

DR and BDR addresses



The protocol of the IPv4 header 0x59 – OSPF messages have a protocol number of 89, and 0x59 means hexadecimal 5 9, which is equal to 89 in decimal, indicating that an OSPF message is encapsulated in this header.